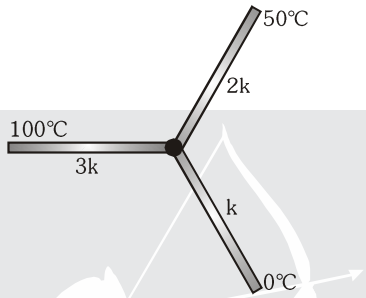
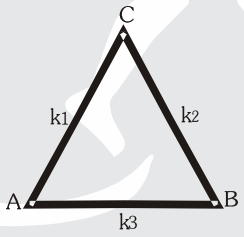
SYLLABUS : HEAT TRANSFER + ALTERNATING CURRENT

1. Three rods of the same dimensions have thermal conductivities 3k, 2k and k. They are arranged as shown, with their ends at 100°C, 50°C and 0°C. The temperature of their junction is:–



(A) 75°C (B) $\frac{200}{3}$°C (C) 40°C (D) $\frac{100}{3}$°C

2. Three rods of same dimensions are arranged as shown in the figure. They have thermal conductivities k_1, k_2 & k_3. The points A and B are maintained at different temperatures. For the heat to flow at the same rate along ACB and AB :–



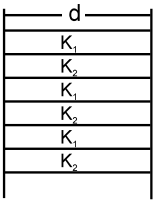
(A) $k_3 = 2(k_1 + k_2)$ (B) $k_3 = \frac{k_1 k_2}{k_1 + k_2}$ (C) $k_3 = k_1 + k_2$ (D) $k_3 = \frac{1}{2}(k_1 + k_2)$

3. A wall has two layers A and B, each made of different material. Both the layers have the same thickness. The thermal conductivity for A is twice that of B. Under steady state, the temperature difference across the whole wall is 36°C. Then the temperature difference across the layer A is

(A) 6°C (B) 12°C (C) 18°C (D) 24°C

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4. A wall consists of alternating blocks with length 'd' and coefficient of thermal conductivity k_1 and k_2. The cross sectional area of the blocks are the same. The equivalent coefficient of thermal conductivity of the wall between left and right is :-



(A) $K_1 + K_2$ (B) $\frac{(K_1+K_2)}{2}$ (C) $\frac{K_1K_2}{K_1+K_2}$ (D) $\frac{2K_1K_2}{K_1+K_2}$

5. A lake surface is exposed to an atmosphere where the temperature is < 0°C. If the thickness of the ice layer formed on the surface grows from 2 cm to 4 cm in 1 hour, The atmospheric temperature, T_a will be-

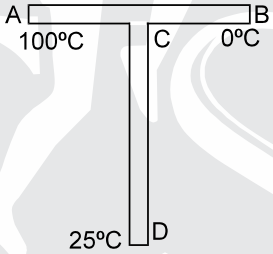
(Thermal conductivity of ice K = 4 x 10^{-3} cal/cm/s/°C; density of ice = 0.9 gm/cc. Latent heat of fusion of ice = 80 cal/gm. Neglect the change of density during the state change. Assume that the water below the ice has 0º temperature every where)

(A) – 20 °C (B) 0 °C (C) – 30 °C (D) – 15 °C

6. A uniform slab of dimension 10cm × 10cm × 1cm is kept between two heat reservoirs at temperatures 10ºC and 90ºC. The larger surface areas touch the reservoirs. The thermal conductivity of the material is 0.80 W/m–ºC. Find the amount of heat flowing through the slab per second.

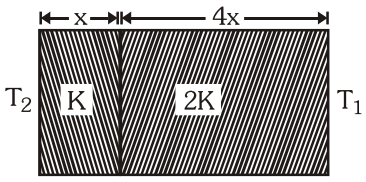
(A) 15 J (B) 32 J (C) 48 J (D) 64 J

7. A rod CD of thermal resistance 5.0 K/W is joined at the middle of an identical rod AB as shown in figure. The ends A, B and D are maintained at 100ºC, 0ºC and 25ºC respectively. Find the heat current in CD.



(A) 2.0 W (B) 4.0 W (C) 6.0 W (D) 8.0 W

8. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x, respectively are T_2 and T_1($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2-T_1)K}{x}\right)f$, with f equals to:–



(A) 1 (B) 1/2 (C) 2/3 (D) 1/3

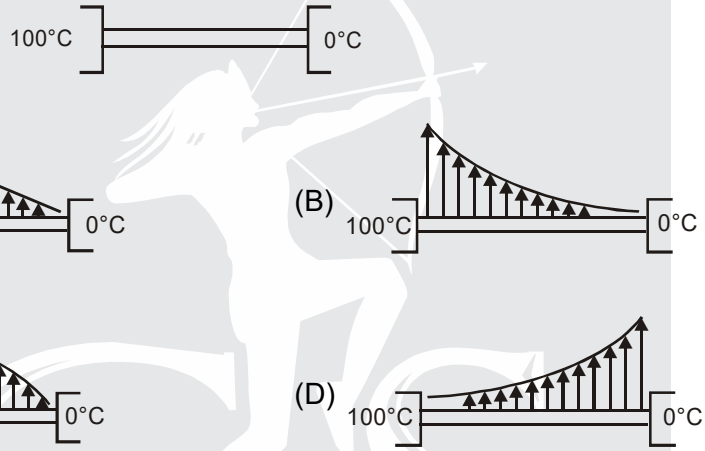
9. Heat flows radially outward through a spherical shell of outside radius R_2 and inner radius R_1. The temperature of inner surface of shell is θ_1 and that of outer is θ_2. The radial distance from centre of shell where the temperature is just half way between θ_1 and θ_2 is :

(A) $\frac{R_1+R_2}{2}$ (B) $\frac{R_1R_2}{R_1+R_2}$ (C) $\frac{2R_1R_2}{R_1+R_2}$ (D) $R_1 + \frac{R_2}{2}$

10. A hollow metallic sphere of radius 20 cm surrounds a concentric metallic sphere of radius 5 cm. The space between the two spheres is filled with a nonmetallic material. The inner and outer spheres are maintained at 50°C and 10°C respectively and it is found that 160 π Joule of heat passes radially from the inner sphere to the outer sphere per second. Find the thermal conductivity of the material between the spheres.

(A) 5 W/m–°C (B) 10 W/m–°C (C) 15 W/m–°C (D) 20 W/m–°C

11. A conducting cylindrical rod of uniform cross-sectional area is kept between two large chambers which are at temperatures 100°C and 0°C respectively. The conductivity of the rod increases with x, where x is distance from 100°C end. The temperature profile of the rod in steady-state will be as :



12. A metal rod of cross-sectional area 1.0 cm^2 is being heated at one end. At one time, the temperature gradient is 5.0°C/cm at cross-section A and is 2.6 °C/cm at cross-section B. Calculate the rate at which the temperature is increasing in the part AB of the rod. The heat capacity of the part AB = 0.40 J/°C, thermal conductivity of the material of the rod = 200 W/m–°C. Neglect any loss of heat to the atmosphere.

(A) 10 °C/s (B) 12 °C/s (C) 5 °C/s (D) 25 °C/s

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13. Assume transmitivity $t \to 0$ for all the cases :

(A) bad absorber is bad emitter (B) bad absorber is good reflector

(C) bad reflector is good emitter (D) bad emitter is good absorber

14. A metallic sphere having radius 0.08 m and mass m = 10kg is heated to a temperature of 227°C and suspended inside a box whose walls are at a temperature of 27°C. The maximum rate at which its temperature will fall is :-

(Take e = 1, Stefan's constant $\sigma = 5.8 \times 10^{-8}$ Wm^{-2} K^{-4} and specific heat of the metal s = 90 cal/kg/deg J = 4.2 Joules/Calorie)

(A) .055 °C/sec (B) .066 °C/sec (C) .044 °C/sec (D) 0.03 °C/sec

15. When q_1 joules of radiation is incident on a body it reflects and transmits total of q_2 joules. Find the emissivity of the body.

(A) $\frac{q_1 + q_2}{q_1}$ (B) $\frac{q_1 - q_2}{2q_1}$ (C) $\frac{q_1 - q_2}{q_1}$ (D) None of these

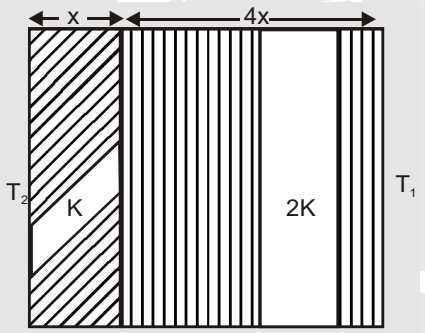
16. Estimate the temperature at which a body may appear blue or red. The values of λ_{mean} for these are 5000 and 7500 Å respectively. [Given Wein's constant b = 0.3 cm K]

(A) 6×10^3 K; 4×10^3K (B) 4×10^3 K; 5×10^3K (C) 3×10^3 K; 4×10^3K (D) None of these

17. The earth receives at its surface radiation from the sun at the rate of 1400 Wm^{-2}. The distance of the centre of the sun from the surface of the earth is 1.5×10^{11} m and the radius of the sun is 7×10^8 m. Treating the sun as a black body calculate temperature of sun.

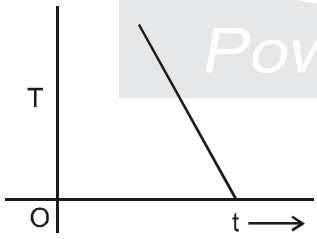
(A) 3803 (B) 4813 (C) 5815 (D) 5803

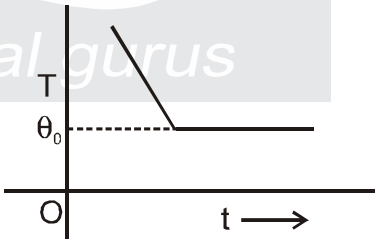
18. The temperature of the two outer surfaces of a composite slab, consisting of two materials K and 2K and thickness x and 4x, respectively, and T_2 and $T_1(T_2 > T_1)$. The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)f$ with f equal to–

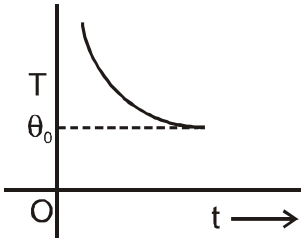


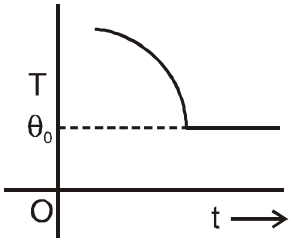
(A) 1 (B) 1/2 (C) 2/3 (D) 1/3

19. If a piece of metal is heated to temperature θ and then allowed to cool ina room which is at temperature θ_0, the graph between the temperature T of the metal and time t will be closest to :

(A)


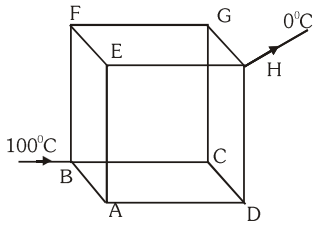
(B)


(C)


(D)


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20. Twelve conducting rods form the sides of a uniform cube of side ℓ. If in steady state, B and H ends of the cube are at 100^0C and 0^0C respectively. Find the temperature of the junction 'A' :–



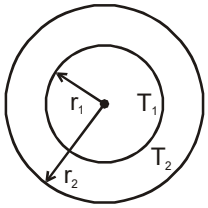
(A) 80^0C (B) 60^0C

(C) 40^0C (D) 70^0C

21. If the termpreature of the sun were to increase from T to 2T and its radius from R to 2R, then the ratio of the radiant energy received on earth to what it was previously will be-

(A) 4 (B) 16 (C) 32 (D) 64

22. The figure shows a system of two concentric spheres of radii r_1 and r_2 and kept at temperature T_1 and T_2, respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to :

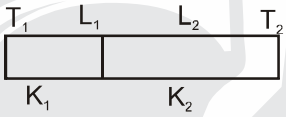


(A) $\frac{(r_2 - r_1)}{(r_1 r_2)}$ (B) $\ln\frac{(r_2)}{(r_1)}$ (C) $\frac{r_1 r_2}{(r_2 - r_1)}$ (D) $(r_2 - r_1)$

23. Assuming the sun to be a spherical body of radius R at a temperature of T K, evaluate the total radiant power, incident on Earth, at a distance r from the Sun. (earth radius = r_0)

(A) $\frac{R^2\sigma T^4}{r^2}$ (B) $\frac{4\pi r_0^2 R^2\sigma T^4}{r^2}$ (C) $\frac{\pi r_0^2 R^2\sigma T^4}{r^2}$ (D) $\frac{r_0^2 R^2\sigma T^4}{4\pi r^2}$

24. One end of a thermally insulated rod is kept at a temperature T_1 and the other at T_2. The rod is composed of two sections of lengths L_1 and L_2 and thermal conductivities k_1 and k_2 respectively. The temperature at the interface of the sections is



(A) $\frac{(K_2 L_2 T_1 + K_1 L_1 T_2)}{(K_1 L_1 + K_2 L_2)}$ (B) $\frac{(K_2 L_1 T_1 + K_1 L_2 T_2)}{(K_2 L_1 + K_1 L_2)}$ (C) $\frac{(K_1 L_2 T_1 + K_2 L_1 T_2)}{(K_1 L_2 + K_2 L_1)}$ (D) $\frac{(K_1 L_1 T_1 + K_2 L_2 T_2)}{(K_1 L_1 + K_2 L_2)}$

25. A spherical solid black body of radius 'r' radiates power 'H' and its rate of cooling is 'C'. If density is constant then which of the following is/are true.

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(A) $H \propto r$ and $c \propto r^2$ (B) $H \propto r^2$ and $c \propto \frac{1}{r}$ (C) $H \propto r$ and $c \propto \frac{1}{r^2}$ (D) $H \propto r^2$ and $c \propto r^2$

ANSWER KEY

1.	(B)	**2.**	(B)	**3.**	(B)	**4.**	(B)	**5.**	(C)
6.	(D)	**7.**	(B)	**8.**	(D)	**9.**	(C)	**10.**	(C)
11.	(B)	**12.**	(B)	**13.**	(ABC)	**14.**	(B)	**15.**	(C)
16.	(A)	**17.**	(D)	**18.**	(D)	**19.**	(C)	**20.**	(B)
21.	(D)	**22.**	(C)	**23.**	(C)	**24.**	(C)	**25.**	(B)